

Call for Expressions of Interest

<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	P ³ : Practical Presentation Planning
<i>Company:</i>	n/a

Contact Details

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Additional Info

<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☐ If Applicable: false
<i>Relevant research areas (if applicable)¹:</i>	Click or tap here to enter text.

Project Background Information

Through this project, you will gain a better understanding of how to discover what potential users want from an application, by developing P³, a practical presentation planning application.

Academia is full of presentations. We ask students to give presentations and we get asked to give presentations too, at events like academic conferences. Planning how to group a set of presentations into 'sessions', where the breaks should be, how long each presentation 'slot' should (or can) be is tricky. It requires some logical thinking (e.g. "if each presentation slot is 20 minutes then I can only fit three into an hour") and some human intuition (e.g. "I think that we need at least a 15 minute break after each session").

Planning presentations is something I find time consuming, but I want to make sure I get it right; P³ should support the process of planning presentations and comparing different schedules with different constraints. The team will need to elicit user needs from potential users of this application (potential users could be members of staff in the School of Computer Science who have to plan presentations, but there are possibly other potential users too) and specify the applications requirements and design. Elicitation could be via interviews or surveys, or a combination of the two; the team should also search for

¹ An overview of different research areas in the school of computer science can be found [here](#).

information about similar scheduling systems ('opposition research') to understand what might already be available. A very good project will validate the requirements/design with the potential users before development of those features, and write them using the Easy Approach to Requirements Syntax (EARS) templates (see <https://alistairmavin.com/ears/> and <https://ieeexplore.ieee.org/abstract/document/5328509>) .

The implementation of P^3 should use constraint programming to provide possible arrangements of presentations, given a set of constraints (e.g. a subset of the presentations must be together, a presentation has to be within a given time window, no more than n presentations in one session, each presentation lasts t minutes, etc.) The User Interface should enable the user to provide whichever constraints they have, view the possible schedules, and compare them (definitely on screen but possibly printed as well).

Vision Statement

For a software project, a vision statement is a succinct description of what the software product should do and who it is going to support.

For people who are planning sets of presentations with a confusing amount of constraints, P^3 is a planning support application that enables the user compare different possible schedules of presentations (possibly with different constraints) to choose which one they prefer. Unlike planning using a white board or a spreadsheet, P^3 enables to user to input and tweak different constraints to see what schedules are possible, for example "can I have 30 minute presentation slots and fit all the presentations in one working day?" Unlike using an LLM, P^3 will use a logical approach to produce possible schedules and leave to the user to make the decisions.

Requirements

The group must provide:

A well designed software product, with a desktop interface, that supports users in comparing different possible schedules of presentations. The software should be supportive, enabling the user to visualise these different possible schedules and decide for themselves which is best. The schedules should be produced using a trustworthy, logical approach. The schedules should be exportable to useful formats (for example CSV, markdown, HTML) so that the schedule can be printed, listed on a webpage, etc. The features of the application should be based on needs elicited from potential users. For example, what sorts of constraints do people consider when trying to plan a set of presentations?

Skills	
<i>Highly desirable:</i> • Requirements Elicitation and Specification • Interviewing or Surveys	<i>Desirable:</i> • Constraint programming or other logical approaches to scheduling • User Interface design
Submission / Award information	
<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.